Week 9 — Risk Measures: Mean–Variance, VaR, CVaR

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Oct 24, 2026

1. Optimization under uncertainty

Deterministic optimization

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & f_i(x) \le 0 \qquad \forall i \in [I] \end{aligned}$$

OUU: uncertainty is captured in the vector $\xi \in \mathbb{R}^k$

objective: $f_0(x, \xi)$ constraints: $f_i(x, \xi)$ $\mathcal{X}$: set of *deterministic* constraints on x

Examples

1) **Portfolio optimization:** ξ = vector of asset returns

$f_0(x, \xi) = -\xi^\top x$ loss $\mathcal{X} = \{ x \in \mathbb{R}^N_+ \ : \ e^\top x = W \}$

(e = all-ones vector, $e^\top x = x_1 + \cdots + x_N$, W = wealth)

2) **Shortest path:** ξ = vector of edge weights

$f_0(x, \xi) = \xi^\top x$, x binary vector $\mathcal{X}$ = flow conservation constraints

Competing paradigms for OUU

Stochastic optimization

$$\min_{x \in \mathcal{X}} \ \mathbb{E}_{\mathbb{P}}\big[f_0(x, \tilde{\xi}) \big]$$

— $\tilde{\xi}$ random, with **known** distribution $\mathbb{P}$

Robust optimization

$$\min_{x \in \mathcal{X}} \ \max_{\xi \in \Xi} \ f_0(x, \xi)$$

— a game against nature: you choose x, nature picks the worst ξ in a set Ξ

↘ ↙

Distributionally robust optimization

$$\min_{x \in \mathcal{X}} \ \max_{\mathbb{Q} \in \mathcal{P}_\lambda(\hat{\mathbb{P}}_N)} \ \mathbb{E}_{\mathbb{Q}}\big[f_0(x, \tilde{\xi}) \big]$$

$\hat{\mathbb{P}}_N$ = empirical distribution of N samples, $\mathcal{P}_\lambda(\hat{\mathbb{P}}_N)$ = ambiguity set: distributions within distance λ of it

— more realistic: don't assume a distribution $\mathbb{P}$; only the data $\hat{\mathbb{P}}_N$

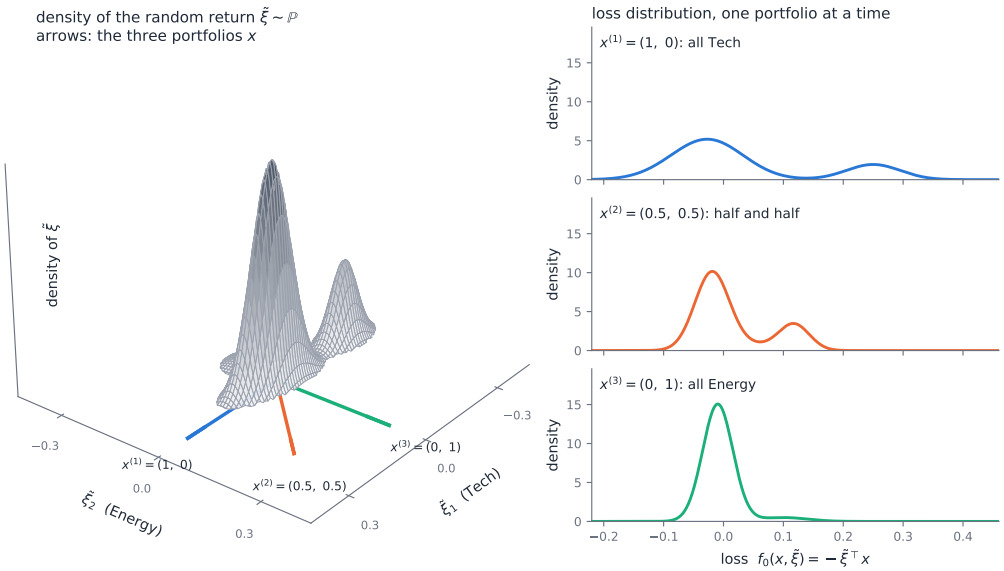
— less conservative: the worst-case *distribution* $\mathbb{Q}$ rather than the worst-case *scenario* ξ

2. Stochastic optimization

$\tilde{\xi} \sim \mathbb{P}$ random *vector*; $\tilde{Z} = f_0(x, \tilde{\xi})$ the loss, a univariate random variable

fix x $\Rightarrow$ $\tilde{Z}$ has a distribution; change x $\Rightarrow$ it changes — **the decision shapes the distribution**

Example (Portfolio optimization)



two assets, Tech (boom/bust) and Energy (quiet); same $\mathbb{P}$ on the left, three portfolios on the right

how do we compare the three? we need one number per distribution.

Def (risk measure) $\mathcal{R}_\mathbb{P} : \mathcal{L} \longrightarrow \mathbb{R}$ $\mathcal{L}$ = all univariate random variables, $\mathbb{R}$ = real numbers

Stochastic program

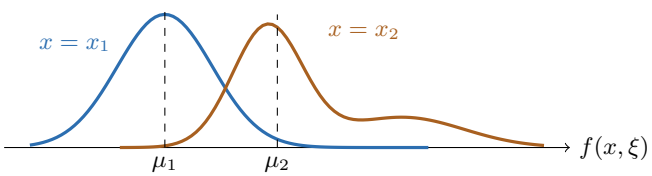
$$\boxed{\begin{array}{lll} \min & \mathcal{R}_\mathbb{P}\big[f_0(x, \tilde{\xi})\big] & \\ \text{s.t.} & x \in \mathcal{X} & \\ & \mathcal{R}_\mathbb{P}\big[f_i(x, \tilde{\xi})\big] \leq 0 & \forall i \in [I] \end{array}} \qquad \text{(SP)}$$

the index $\mathbb{P}$ is the distribution of $\tilde{\xi}$: every risk measure depends on it

Examples of risk measures

Risk neutral $\mathcal{R}_\mathbb{P} = \mathbb{E}_\mathbb{P}$: expected value

$$\begin{array}{lll} \min & \mathbb{E}_\mathbb{P}\big[f_0(x, \tilde{\xi})\big] & \\ \text{s.t.} & x \in \mathcal{X} & \\ & \mathbb{E}_\mathbb{P}\big[f_i(x, \tilde{\xi})\big] \leq 0 & \forall i \in [I] \end{array}$$



$\mu_1 = \mathbb{E}_\mathbb{P}[f_0(x_1, \tilde{\xi})] < \mu_2 = \mathbb{E}_\mathbb{P}[f_0(x_2, \tilde{\xi})]$ $\Rightarrow$ x_1 is preferable under the risk-neutral measure.

3. Risk averse 1: mean–variance

Risk averse 1) **mean–variance** $\mathcal{R}_{\mathbb{P}} = \lambda \operatorname{Var}_{\mathbb{P}} + (1-\lambda)\,\mathbb{E}_{\mathbb{P}}$: convex combination of mean and variance

$$\min \quad \lambda \operatorname{Var}_{\mathbb{P}}\big[f_0(x,\tilde{\xi})\big] + (1-\lambda)\,\mathbb{E}_{\mathbb{P}}\big[f_0(x,\tilde{\xi})\big] \qquad \lambda \in [0,1]$$

Example: mean–variance portfolio optimization (Harry Markowitz, 1952)

$\tilde{\xi} \in \mathbb{R}^N$ is the vector of asset returns. Assume we know

$\mathbb{E}_{\mathbb{P}}[\tilde{\xi}] = \mu$ vector of means, $\mathbb{E}_{\mathbb{P}}\big[(\tilde{\xi}-\mu)(\tilde{\xi}-\mu)^\top\big] = \mathbb{E}_{\mathbb{P}}[\tilde{\xi}\tilde{\xi}^\top] - \mu\mu^\top = \Sigma$ covariance matrix

an investor allocates a wealth of \$1 to minimize the mean–variance of the portfolio loss $-\tilde{\xi}^\top x$:

$$\begin{array}{ll} \min & \lambda \operatorname{Var}_{\mathbb{P}}\big(-\tilde{\xi}^\top x\big) + (1-\lambda)\,\mathbb{E}_{\mathbb{P}}\big[-\tilde{\xi}^\top x\big] \\ \text{s.t.} & x \in \mathbb{R}^N_+ \\ & e^\top x = 1 \end{array}$$

reformulate — both terms are functions of μ and Σ only:

$\mathbb{E}_{\mathbb{P}}[-\tilde{\xi}^\top x] = -\mu^\top x$ linearity of expectation

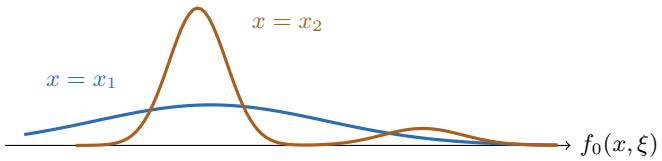
$$\begin{aligned} \operatorname{Var}_{\mathbb{P}}(-\tilde{\xi}^\top x) = \operatorname{Var}_{\mathbb{P}}(\tilde{\xi}^\top x) &= \mathbb{E}_{\mathbb{P}}\big[(\tilde{\xi}^\top x)^2\big] - \big(\mathbb{E}_{\mathbb{P}}[\tilde{\xi}^\top x]\big)^2 \\ &= \mathbb{E}_{\mathbb{P}}\big[x^\top \tilde{\xi}\tilde{\xi}^\top x\big] - (\mu^\top x)^2 = x^\top \mathbb{E}_{\mathbb{P}}[\tilde{\xi}\tilde{\xi}^\top]\, x - x^\top \mu\mu^\top x \\ &= x^\top \big(\mathbb{E}_{\mathbb{P}}[\tilde{\xi}\tilde{\xi}^\top] - \mu\mu^\top\big) x \\ &= x^\top \Sigma\, x \end{aligned}$$

$$\begin{array}{ll} \min & \lambda\, x^\top \Sigma\, x - (1-\lambda)\,\mu^\top x \\ \text{s.t.} & x \in \mathbb{R}^N_+ \\ & e^\top x = 1 \end{array}$$

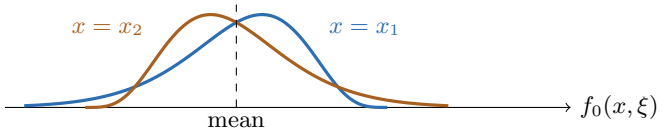
$\Sigma \succeq 0 \;\Rightarrow$ convex quadratic program

Limits of mean–variance

- for a general loss $f_0(x,\tilde{\xi})$, $\operatorname{Var}_{\mathbb{P}}[f_0(x,\tilde{\xi})]$ need not be convex in x — the mean–variance risk measure in general leads to a non-convex optimization problem. (linear loss: convex, as above)
- variance penalizes spread in *both* directions — it does not see where the big losses are:



x_2 has lower variance, but has higher probability of generating bigger losses.

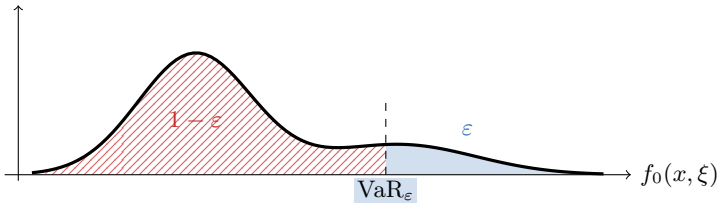


x_1 and x_2 have the same mean and the same variance (mirror images), so mean–variance cannot tell them apart; but x_1 is preferred because it has higher probability of generating smaller losses.

4. Risk averse 2: Value at Risk (VaR)

2) **Value at Risk** $\mathcal{R}_{\mathbb{P}} = \mathrm{VaR}_\varepsilon$: $(1-\varepsilon)$-quantile of the loss typically a non-convex optimization problem

Of course, VaR_ε depends on $\mathbb{P}$ just like $\mathbb{E}_{\mathbb{P}}$ and $\mathrm{Var}_{\mathbb{P}}$; we omit the index for notational simplicity.



— popular in finance

— imposed by the Basel II accord: how much capital banks need to set aside to hold against financial risks

Formally:

$$\mathrm{VaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] \;=\; \min \quad t \qquad \text{s.t.} \quad \mathbb{P}\big(f_0(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon$$

← chance / probability constraint

generally non-convex, highly intractable

the smallest loss level t that is exceeded with probability at most ε

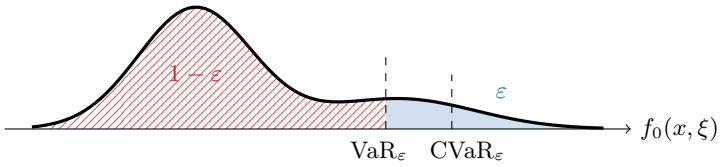
VaR risk measure in general leads to a non-convex optimization problem.

5. Risk averse 3: Conditional Value at Risk (CVaR)

3) **Conditional Value at Risk** $\mathcal{R}_{\mathbb{P}} = \mathrm{CVaR}_\varepsilon$: average loss in the worst ε fraction of outcomes (Rockafellar & Uryasev, 2000)

— CVaR leads to a **convex** optimization problem if $f(x,\xi)$ is convex in x

— imposed by the Basel III accord



$$\mathrm{CVaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] \;=\; \min_{\beta\in\mathbb{R}} \;\; \beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_0(x,\tilde{\xi})-\beta,\;0\big\}\Big]$$

the objective is convex in (x,β) when f_0 is convex in x: $\max\{\text{convex},0\}$ is convex, and $\mathbb{E}$ keeps it convex.

Recall $\tilde{Z} = f_0(x,\tilde{\xi})$.

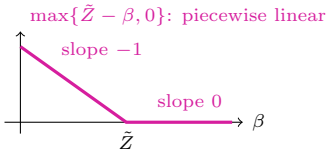
Proposition: if $\tilde{Z}$ has a continuous distribution, then $\mathrm{CVaR}_\varepsilon[\tilde{Z}] = \mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z} \ge \mathrm{VaR}_\varepsilon[\tilde{Z}]\big]$

— CVaR = the **average loss in the worst** ε **fraction** of outcomes; so $\mathrm{CVaR}_\varepsilon \ge \mathrm{VaR}_\varepsilon$.

Proof: step 1 — find the minimizer β^*.

$$\frac{\partial}{\partial\beta}\Big(\beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{\tilde{Z}-\beta,0\}\big]\Big) = 0$$

$$1 + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\frac{\partial}{\partial\beta}\max\{\tilde{Z}-\beta,0\}\Big] = 0$$



$$1 + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[-\mathbb{1}_{\tilde{Z}>\beta}\big] = 0 \qquad \text{indicator function: } \mathbb{1}_A = \begin{cases} 1 & \text{if } A \text{ holds} \\ 0 & \text{otherwise} \end{cases}$$

$$1 - \frac{1}{\varepsilon}\,\mathbb{P}(\tilde{Z}>\beta) = 0 \qquad \mathbb{E}_{\mathbb{P}}[\mathbb{1}_A] = \mathbb{P}(A)$$

$$\mathbb{P}(\tilde{Z}>\beta) = \varepsilon \qquad \Rightarrow \qquad \beta^* = \mathrm{VaR}_\varepsilon[\tilde{Z}]$$

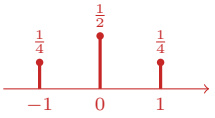
step 2 — plug β^* in. Since $\tilde{Z}$ is continuous, $\mathbb{P}(\tilde{Z}\ge\beta^*) = \mathbb{P}(\tilde{Z}>\beta^*) = \varepsilon$:

$$\begin{aligned}
\mathrm{CVaR}_\varepsilon[\tilde{Z}] &= \beta^* + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{\tilde{Z}-\beta^*,0\}\big] \\
&= \beta^* + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[(\tilde{Z}-\beta^*)\,\mathbb{1}_{\tilde{Z}\ge\beta^*}\big] \\
&= \beta^* + \frac{1}{\mathbb{P}(\tilde{Z}\ge\beta^*)}\,\mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\mathbb{1}_{\tilde{Z}\ge\beta^*}\big] - \frac{1}{\mathbb{P}(\tilde{Z}\ge\beta^*)}\,\mathbb{E}_{\mathbb{P}}\big[\beta^*\,\mathbb{1}_{\tilde{Z}\ge\beta^*}\big] \\
&= \beta^* + \mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z}\ge\beta^*\big] - \beta^* \\
&= \mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z}\ge\mathrm{VaR}_\varepsilon[\tilde{Z}]\big] \qquad \blacksquare
\end{aligned}$$

> **Def (conditional expectation)** $\qquad \mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z}\in A\big] = \dfrac{\mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\mathbb{1}_{\tilde{Z}\in A}\big]}{\mathbb{P}(\tilde{Z}\in A)}$

$\tilde{Z}$ has a discrete distribution — what does the proposition become?

Consider: $\mathbb{P}(\tilde{Z}=1)=\frac14$, $\mathbb{P}(\tilde{Z}=0)=\frac12$, $\mathbb{P}(\tilde{Z}=-1)=\frac14$. Let $\varepsilon=\frac12$.



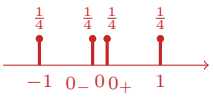
$$\mathrm{VaR}_\varepsilon[\tilde{Z}] = \min_{t\in\mathbb{R}}\ t \qquad\qquad = 0 \qquad (\mathbb{P}(\tilde{Z}\le 0) = \tfrac34 \ge \tfrac12,\ \text{and for any } t<0,\ \mathbb{P}(\tilde{Z}\le t)\le\tfrac14)$$
$$\text{s.t. } \mathbb{P}(\tilde{Z}\le t) \ge 1-\tfrac12$$

$$\begin{aligned}
\mathrm{CVaR}_\varepsilon[\tilde{Z}] &= \min_{\beta\in\mathbb{R}}\ \beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{\tilde{Z}-\beta,0\}\big] \\
&= \min_{\beta\in\mathbb{R}}\ \beta + \frac{1}{1/2}\Big(\tfrac14\max\{1-\beta,0\} + \tfrac12\max\{0-\beta,0\} + \tfrac14\max\{-1-\beta,0\}\Big) \\
&= \tfrac12 \qquad (\text{attained at } \beta=0\text{: the objective is } \tfrac12+\tfrac{\beta}{2} \text{ on } [0,1] \text{ and } \tfrac12-\tfrac{\beta}{2} \text{ on } [-1,0])
\end{aligned}$$

but $\quad \mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z}\ge\underbrace{\mathrm{VaR}_\varepsilon[\tilde{Z}]}_{0}\big] = \dfrac{\frac12\times0+\frac14\times1}{\frac12+\frac14} = \frac13$ $\qquad$ and $\qquad \mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z}>\mathrm{VaR}_\varepsilon[\tilde{Z}]\big] = \dfrac{\frac14\times1}{\frac14} = 1$

neither is $\frac12$: the atom at VaR_ε is too big — $\mathbb{P}(\tilde{Z}\ge0)=\frac34>\varepsilon$ but $\mathbb{P}(\tilde{Z}>0)=\frac14<\varepsilon$.

Atom splitting split the atom at 0 so that exactly $\varepsilon=\frac12$ of the mass lies in the tail:



$$\mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\big|\,\tilde{Z}\ge0_+\big] = \frac{\frac14\times0+\frac14\times1}{\frac12} = \frac12 \qquad = \mathrm{CVaR}_\varepsilon[\tilde{Z}] \quad \checkmark$$

in general: CVaR = the average of the worst ε of the probability mass, with the atom at VaR_ε split as needed. The $\min_\beta$ formula gets this right automatically.

6. Uncertainty in the constraint

How to handle uncertainty in the constraint? $\mathcal{R}_{\mathbb{P}}\big[f_i(x,\tilde{\xi})\big] \le 0$

— $\mathbb{E}_{\mathbb{P}}\big[f_i(x,\tilde{\xi})\big] \le 0$ the constraint holds *on average*

— $\mathrm{VaR}_\varepsilon\big[f_i(x,\tilde{\xi})\big] \le 0$ non-convex in general

$\iff \min\limits_{t\in\mathbb{R}} \big\{\, t \;:\; \mathbb{P}\big(f_i(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon \,\big\} \le 0$

$\iff \exists\, t \le 0 \;:\; \mathbb{P}\big(f_i(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon$

$\iff \mathbb{P}\big(f_i(x,\tilde{\xi}) \le 0\big) \ge 1-\varepsilon$

— $\mathrm{CVaR}_\varepsilon\big[f_i(x,\tilde{\xi})\big] \le 0$ convex; gives an upper bound on VaR_ε, so it implies the VaR constraint

$\iff \min\limits_{\beta}\; \beta + \dfrac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{f_i(x,\tilde{\xi})-\beta, 0\}\big] \le 0$

$\iff \exists\, \beta\in\mathbb{R} \;:\; \beta + \dfrac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{f_i(x,\tilde{\xi})-\beta, 0\}\big] \le 0$

Chance constraint / probabilistic constraint $\mathbb{P}\big(f_i(x,\tilde{\xi}) \le 0\big) \ge 1-\varepsilon$

> **Proposition:** VaR_ε constraint $\iff$ chance constraint

Putting it together — the CVaR stochastic program

one β per risk term: β_0 for the objective, β_i for constraint i; all of them join the outer min

$$\begin{aligned} \min_{x\in\mathcal{X},\ \beta_0,\dots,\beta_I\in\mathbb{R}} \quad & \beta_0 + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_0(x,\tilde{\xi})-\beta_0,\ 0\big\}\Big] \\ \text{s.t.} \quad & \beta_i + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_i(x,\tilde{\xi})-\beta_i,\ 0\big\}\Big] \le 0 \qquad \forall i\in[I] \end{aligned}$$

convex whenever every f_i is convex in x.

with S scenarios ξ^s of probabilities p_s (i.i.d. data: $p_s = \frac{1}{S}$), replace each max by a variable $u_{is} \ge 0$

$$\begin{aligned} \min \quad & \beta_0 + \frac{1}{\varepsilon}\sum_{s\in[S]} p_s\, u_{0s} \\ \text{s.t.} \quad & \beta_i + \frac{1}{\varepsilon}\sum_{s\in[S]} p_s\, u_{is} \le 0 \qquad\qquad \forall i\in[I] \\ & u_{is} \ge f_i(x,\xi^s) - \beta_i, \quad u_{is} \ge 0 \quad \forall i\in\{0\}\cup[I],\ s\in[S] \\ & x\in\mathcal{X} \end{aligned}$$

linear f_i and polyhedral $\mathcal{X}$ $\Rightarrow$ a linear program. The same trick on VaR_ε needs a binary variable per scenario (which scenarios may violate the constraint) — a mixed-integer program. That is why CVaR is the tractable one.